



6133 - Identifying the progenitors of a complete sample of long gamma-ray bursts

Cycle: 3, Proposal Category: GO

INVESTIGATORS

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Name	Institution
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Dr. Matt Nicholl (CoI) (ESA Member)	Queen's University Belfast
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Dr. Pall Jakobsson (CoI)	University of Iceland
Dr. Ruben Salvaterra (CoI) (ESA Member)	INAF - IASF Milano
Sam Oates (CoI) (ESA Member)	Lancaster University
Dr. Tanmoy Laskar (CoI)	University of Utah
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Prof. Ilya Mandel (CoI)	Monash University

OBSERVATIONS

Folder	Observation	Label	Observing Template	Science Target
Observation Folder				
	1		NIRSpec Fixed Slit Spectroscopy	(1) GRB240825A
	2		NIRSpec Fixed Slit Spectroscopy	(2) GRB241001A
	3		NIRSpec Fixed Slit Spectroscopy	(3) GRB241010A
	4		NIRSpec Fixed Slit Spectroscopy	(4) EP250108a
	5		NIRSpec Fixed Slit Spectroscopy	(5) GRB250419A
	6		NIRSpec Fixed Slit Spectroscopy	(6) GRB250424A
	7		NIRSpec Fixed Slit Spectroscopy	(7) GRB250625A

ABSTRACT

The discovery of kilonovae alongside gamma-ray bursts (GRBs) lasting tens of seconds or more has sparked a paradigm shift in our understanding of the diversity of jets launched by merging neutron stars. The prototypes of this new class are two of the brightest GRBs ever detected, strongly suggesting a hidden population that are systematically misclassified as collapsing massive stars - the canonical model for GRBs that last for more than two seconds. Long-lived mergers have palpable implications for binary neutron star merger rates, heavy element nucleosynthesis, the physics of relativistic jet launch, and multi-messenger follow-up strategies to gravitational wave events. We propose NIRSpec observations over the next three JWST cycles to systematically target and classify 15 emergent GRBs at $z < 1$ as either mergers or collapsars based on spectroscopic identification or exclusion of supernovae. We will build the first 'clean' observational samples of collapsars and mergers to investigate their respective rates and observable properties. For mergers up to $z = 0.5$, we expect to make spectral detections of kilonovae, potentially resulting in the largest spectroscopic sample of this rare class of transient to date. We will also extend the redshift range of detected type Ic broad-lined supernovae, which accompany

long GRBs, out to $z = 1$. This will allow us to uniquely probe their evolution at redshift $0.5 < z < 1$ for the first time, providing insights into how the composition of massive stars and their host galaxies change in this cosmic epoch.

OBSERVING DESCRIPTION

We request NIRSpec observations of 15 well-localised GRBs with confirmed redshifts of up to $z \sim 1$. These are targets of opportunity, and so are not known in advance. Observations will take place over a 3-year period to build unbiased, systematic samples of GRBs from stellar collapse and from long-lived mergers. We aim to spectroscopically detect or rule out supernovae, and detect kilonovae where sufficiently nearby ($z \sim 0.5$ or less). We will use the Prism/CLEAR grating/filter pair and the S200A1 slit (0.2" x 3.8"). The detector setup utilises the full subarray and the NRSIRS2RAPID readout pattern.

Acquisition will be done with WATA using an offset star, but this cannot be planned until the observations are known.

Proposal 6133 - Targets - Identifying the progenitors of a complete sample of long gamma-ray bursts

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	GRB240825A	RA: 22 58 17.2640 (344.5719333d) Dec: +01 01 36.77 (1.02688d) Equinox: J2000 <i>Comments:</i> <i>Category=Star</i> <i>Description=[Gamma Ray bursters]</i> <i>Extended=NO</i>		
	(2)	GRB241001A	RA: 01 22 12.7510 (20.5531292d) Dec: -43 28 31.95 (-43.47554d) Equinox: J2000 <i>Comments:</i> <i>Category=Star</i> <i>Description=[Gamma Ray bursters]</i> <i>Extended=NO</i>		
	(3)	GRB241010A	RA: 10 13 31.5340 (153.3813917d) Dec: +11 32 3.20 (11.53422d) Equinox: J2000 <i>Comments:</i> <i>Category=Star</i> <i>Description=[Gamma Ray bursters]</i> <i>Extended=NO</i>		
	(4)	EP250108a	RA: 03 42 28.3920 (55.6183000d) Dec: -22 30 21.27 (-22.50591d) Equinox: J2000 <i>Comments:</i> <i>Category=Star</i> <i>Description=[Gamma Ray bursters]</i> <i>Extended=NO</i>		
	(5)	GRB250419A	RA: 13 29 37.2870 (202.4053625d) Dec: +07 02 27.86 (7.04107d) Equinox: J2000 <i>Comments:</i> <i>Category=Star</i> <i>Description=[Gamma Ray bursters]</i>		
	(6)	GRB250424A	RA: 14 29 59.9610 (217.4998375d) Dec: -35 01 30.86 (-35.02524d) Equinox: J2000 <i>Comments:</i> <i>Category=Star</i> <i>Description=[Gamma Ray bursters]</i> <i>Extended=NO</i>		
	(7)	GRB250625A	RA: 17 26 2.1200 (261.5088333d) Dec: +22 16 3.61 (22.26767d) Equinox: J2000 <i>Comments:</i> <i>Category=Star</i> <i>Description=[Gamma Ray bursters]</i> <i>Extended=NO</i>		

Proposal 6133 - Targets - Identifying the progenitors of a complete sample of long gamma-ray bursts

(16)	GRB240825A_acq_star	RA: 22 58 18.1550 (344.5756458d) Dec: +01 00 49.13 (1.01365d) Equinox: J2000 <i>Comments:</i> <i>Category=Star</i> <i>Description=[Gamma Ray bursters]</i> <i>Extended=NO</i>
(17)	GRB241001A_acq_star	RA: 01 22 13.5030 (20.5562625d) Dec: -43 29 1.98 (-43.48388d) Equinox: J2000 <i>Comments:</i> <i>Category=Star</i> <i>Description=[Gamma Ray bursters]</i> <i>Extended=NO</i>
(18)	GRB241010A_acq_star	RA: 10 13 30.9380 (153.3789083d) Dec: +11 31 59.89 (11.53330d) Equinox: J2000 <i>Comments:</i> <i>Category=Star</i> <i>Description=[Gamma Ray bursters]</i> <i>Extended=NO</i>
(19)	EP250108a_acq_star	RA: 03 42 28.3330 (55.6180542d) Dec: -22 30 24.69 (-22.50686d) Equinox: J2000 <i>Comments:</i> <i>Category=Star</i> <i>Description=[Gamma Ray bursters]</i> <i>Extended=NO</i>
(20)	GRB250419A_acq_star	RA: 13 29 32.8770 (202.3869875d) Dec: +07 02 15.92 (7.03776d) Equinox: J2000 <i>Comments:</i> <i>Category=Star</i> <i>Description=[Gamma Ray bursters]</i> <i>Extended=NO</i>
(21)	GRB250424A_acq_star	RA: 14 30 1.3040 (217.5054333d) Dec: -35 01 24.69 (-35.02353d) Equinox: J2000 <i>Comments:</i> <i>Category=Star</i> <i>Description=[Gamma Ray bursters]</i> <i>Extended=NO</i>
(22)	GRB250625A_acq_star	RA: 17 26 3.6140 (261.5150583d) Dec: +22 15 55.02 (22.26528d) Equinox: J2000 <i>Comments:</i> <i>Category=Star</i> <i>Description=[Gamma Ray bursters]</i> <i>Extended=NO</i>

Proposal 6133 - Targets - Identifying the progenitors of a complete sample of long gamma-ray bursts

Generic Targets	#	Name	Criteria	Description
	(8)	GRB8	GRB at $z < \sim 1$	
	(9)	GRB9	GRB at $z < \sim 1$	
	(10)	GRB10	GRB at $z < \sim 1$	
	(11)	GRB11	GRB at $z < \sim 1$	
	(12)	GRB12	GRB at $z < \sim 1$	
	(13)	GRB13	GRB at $z < \sim 1$	
	(14)	GRB14	GRB at $z < \sim 1$	
	(15)	GRB15	GRB at $z < \sim 1$	

Proposal 6133 - Observation 1 - Identifying the progenitors of a complete sample of long gamma-ray bursts

Observation	Proposal 6133, Observation 1										Mon Jun 30 23:00:13 GMT 2025	
	Diagnostic Status: Warning											
Observing Template: NIRSpec Fixed Slit Spectroscopy												
Diagnostics	(Observation 1) Warning (Form): The slew between the acquisition exposure and the farthest science exposure is 61.261 Arcsec (larger than the recommended limit of 40.000 Arcsec) and may result in reduced or no schedulability. See more information in the diagnostic browser.											
	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates		Targ. Coord. Corrections				Miscellaneous			
	(1)	GRB240825A	RA: 22 58 17.2640 (344.5719333d) Dec: +01 01 36.77 (1.02688d) Equinox: J2000									
Comments: Category=Star Description=[Gamma Ray bursters] Extended=NO												
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	16 GRB240825A_ac q_star	WATA	SUB2048	F110W	NRSRAPID	3	1	1	3.628	1	
Template	HFF Readout Mode			Slit				Subarray				
	false			S200A1				FULL				
Dithers	#	Primary Dither Positions						Sub-Pixel Pattern				
	1	3						SPATIAL				
Spectral Elements	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	PRISM/CLEAR	S200A1	NRSIRS2RAPID	35	2	1	NONE	6	12	6302.4	177949

Proposal 6133 - Observation 1 - Identifying the progenitors of a complete sample of long gamma-ray bursts

Special Requirements	Before Date 10-NOV-2024:00:00:00 Aperture PA Range 210 to 215 Degrees (V3 71.15809631 to 76.15809631)
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Proposal 6133 - Observation 2 - Identifying the progenitors of a complete sample of long gamma-ray bursts

Observation	Proposal 6133, Observation 2										Mon Jun 30 23:00:13 GMT 2025	
	Diagnostic Status: Warning											
	Observing Template: NIRSspec Fixed Slit Spectroscopy											
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(2)	GRB241001A	RA: 01 22 12.7510 (20.5531292d) Dec: -43 28 31.95 (-43.47554d) Equinox: J2000									
	Comments: Category=Star Description=[Gamma Ray bursters] Extended=NO											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	17 GRB241001A_ac q_star	WATA	SUB2048	F140X	NRSRAPID	3	1	1	3.628	1	
Template	HFF Readout Mode			Slit			Subarray					
	false			S400A1			FULL					
Dithers	#	Primary Dither Positions						Sub-Pixel Pattern				
	1	3						SPATIAL				
Spectral Elements	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	PRISM/CLEAR	S400A1	NRSIRS2RAPID	35	2	1	NONE	6	12	6302.4	177949

Proposal 6133 - Observation 2 - Identifying the progenitors of a complete sample of long gamma-ray bursts

Special Requirements	Between Dates 22-OCT-2024:00:00:00 and 29-OCT-2024:00:00:00
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Proposal 6133 - Observation 3 - Identifying the progenitors of a complete sample of long gamma-ray bursts

Observation	Proposal 6133, Observation 3										Mon Jun 30 23:00:13 GMT 2025	
	Diagnostic Status: Warning											
Observing Template: NIRSpec Fixed Slit Spectroscopy												
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(3)	GRB241010A	RA: 10 13 31.5340 (153.3813917d) Dec: +11 32 3.20 (11.53422d) Equinox: J2000									
Comments: Category=Star Description=[Gamma Ray bursters] Extended=NO												
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	18 GRB241010A_ac q_star	WATA	SUB2048	F140X	NRSRAPID	3	1	1	3.628	215368	
Template	HFF Readout Mode				Slit				Subarray			
	false				S400A1				FULL			
Dithers	#	Primary Dither Positions						Sub-Pixel Pattern				
	1	3						SPATIAL				
Spectral Elements	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	PRISM/CLEAR	S400A1	NRSIRS2RAPID	35	2	1	NONE	6	12	6302.4	177949

Proposal 6133 - Observation 3 - Identifying the progenitors of a complete sample of long gamma-ray bursts

Special Requirements	Between Dates 19-NOV-2024:00:00:00 and 26-NOV-2024:00:00:00
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Proposal 6133 - Observation 4 - Identifying the progenitors of a complete sample of long gamma-ray bursts

Observation	Proposal 6133, Observation 4										Mon Jun 30 23:00:13 GMT 2025	
	Diagnostic Status: Warning											
	Observing Template: NIRSspec Fixed Slit Spectroscopy											
Diagnostics	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(4)	EP250108a	RA: 03 42 28.3920 (55.6183000d) Dec: -22 30 21.27 (-22.50591d) Equinox: J2000									
	Comments: Category=Star Description=[Gamma Ray bursters] Extended=NO											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	19 EP250108a_acq_s tar	WATA	FULL	F110W	NRSRAPID	3	1	1	42.947	1	
Template	HFF Readout Mode				Slit			Subarray				
	false				S400A1			FULL				
Dithers	#	Primary Dither Positions						Sub-Pixel Pattern				
	1	3						SPATIAL				
Spectral Elements	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	PRISM/CLEAR	S400A1	NRSIRS2RAPID	35	2	1	NONE	6	12	6302.4	177949

Proposal 6133 - Observation 4 - Identifying the progenitors of a complete sample of long gamma-ray bursts

Special Requirements	Before Date 02-FEB-2025:00:00:00
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Proposal 6133 - Observation 5 - Identifying the progenitors of a complete sample of long gamma-ray bursts

Observation	Proposal 6133, Observation 5											Mon Jun 30 23:00:13 GMT 2025
	Diagnostic Status: Warning											
	Observing Template: NIRSpec Fixed Slit Spectroscopy											
Diagnostics	(Observation 5) Warning (Form): The slew between the acquisition exposure and the farthest science exposure is 79.548 Arcsec (larger than the recommended limit of 40.000 Arcsec) and may result in reduced or no schedulability. See more information in the diagnostic browser.											
	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(5)	GRB250419A	RA: 13 29 37.2870 (202.4053625d) Dec: +07 02 27.86 (7.04107d) Equinox: J2000									
	Comments: Category=Star Description=[Gamma Ray bursters]											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	20 GRB250419A_ac q_star	WATA	SUB32	CLEAR	NRSRAPID	3	1	1	0.08	1	
Template	HFF Readout Mode				Slit				Subarray			
	false				S200A1				FULL			
Dithers	#	Primary Dither Positions						Sub-Pixel Pattern				
	1	3						SPATIAL				
Spectral Elements	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	PRISM/CLEAR	S200A1	NRSIRS2RAPID	35	2	1	NONE	6	12	6302.4	177949

Special Requirements	Before Date 26-MAY-2025:00:00:00
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Proposal 6133 - Observation 6 - Identifying the progenitors of a complete sample of long gamma-ray bursts

Observation	Proposal 6133, Observation 6										Mon Jun 30 23:00:13 GMT 2025	
	Diagnostic Status: Warning											
	Observing Template: NIRSspec Fixed Slit Spectroscopy											
Diagnostics	(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(6)	GRB250424A	RA: 14 29 59.9610 (217.4998375d) Dec: -35 01 30.86 (-35.02524d) Equinox: J2000									
	Comments: Category=Star Description=[Gamma Ray bursters] Extended=NO											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	21 GRB250424A_ac q_star	WATA	SUB32	CLEAR	NRSRAPIDD6	3	1	1	0.26	1	
Template	HFF Readout Mode				Slit				Subarray			
	false				S200A1				FULL			
Dithers	#	Primary Dither Positions						Sub-Pixel Pattern				
	1	3						SPATIAL				
Spectral Elements	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	PRISM/CLEAR	S200A1	NRSIRS2RAPID	35	2	1	NONE	6	12	6302.4	177949

Proposal 6133 - Observation 6 - Identifying the progenitors of a complete sample of long gamma-ray bursts

Special Requirements	Before Date 25-JUN-2025:00:00:00
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Proposal 6133 - Observation 7 - Identifying the progenitors of a complete sample of long gamma-ray bursts

Observation	Proposal 6133, Observation 7											Mon Jun 30 23:00:13 GMT 2025
	Diagnostic Status: Warning											
	Observing Template: NIRSspec Fixed Slit Spectroscopy											
	Comments: Please schedule as soon as possible.											
Diagnostics	(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(7)	GRB250625A	RA: 17 26 2.1200 (261.5088333d) Dec: +22 16 3.61 (22.26767d) Equinox: J2000									
	Comments: Category=Star Description=[Gamma Ray bursters] Extended=NO											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	22 GRB250625A_ac q_star	WATA	SUB32	CLEAR	NRSRAPIDD6	3	1	1	0.26	1	
Template	HFF Readout Mode				Slit				Subarray			
	false				S400A1				FULL			
Dithers	#	Primary Dither Positions							Sub-Pixel Pattern			
	1	3							SPATIAL			
Spectral Elements	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	PRISM/CLEAR	S400A1	NRSIRS2RAPID	35	2	1	NONE	6	12	6302.4	177949

Proposal 6133 - Observation 7 - Identifying the progenitors of a complete sample of long gamma-ray bursts

Special Requirements	Before Date 15-JUL-2025:00:00:00
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